

5(a)	Progressive: all particles have the same amplitude Stationary: maximum to minimum/zero amplitude Progressive: adjacent particles within one wavelength are not in phase Stationary: waves particles are in phase between adjacent nodes and/or wave particles between adjacent nodes are in antiphase to wave particles in the next adjacent nodes.	[1] [1]
(b)(i)	wavelength = 1.2 m(zero displacement at 0. 0.60 m, 1.2 m, 1.8 m, 2.4 m) either peaks at 0.30 m and 1.5 m and troughs at 0.90 m and 2.1 m or vice versa (but not both) maximum amplitude 5.0 mm	[1] [1]
(ii)	180° or π rad	[1]
(iii)	$t = 0$, particles has KE as particle is moving $t = 5.0$ ms, particle has no KE as particle is stationary so decrease in KE between $t = 0$ to 5.0 ms.	[1] [1]

6(a)(i)	$R = 0 \text{ } / \text{ } A$	
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